



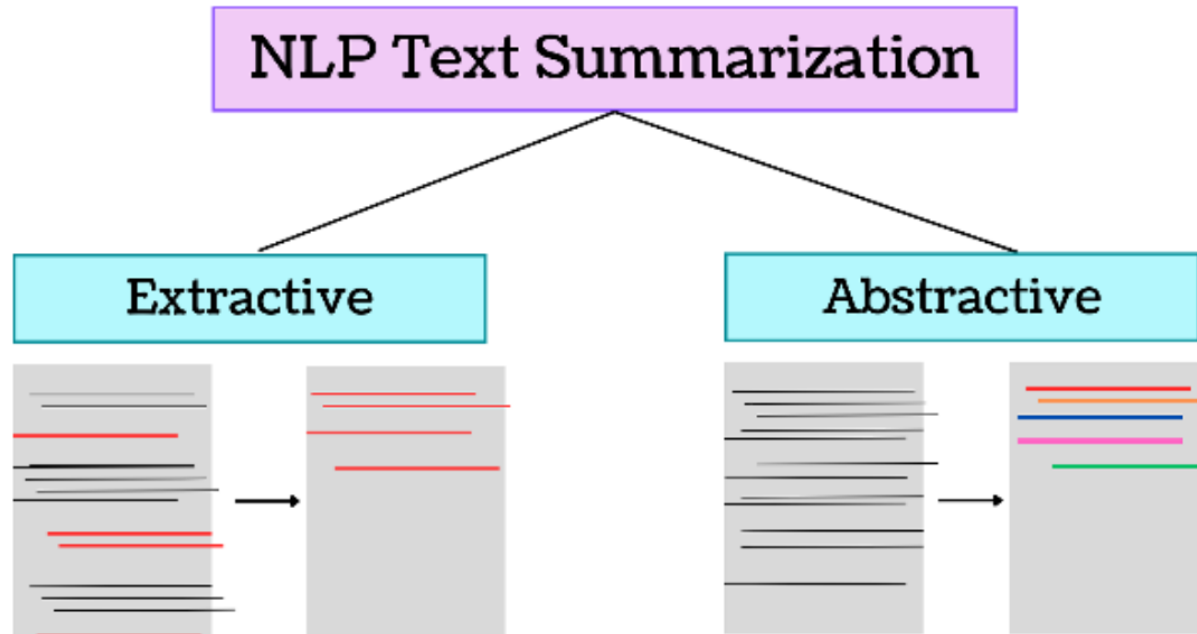
University
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COMM117

Large Language Models and Application

Two main types

- **Extractive:** select sentences/spans from source
- **Abstractive:** generate new wording (paraphrase + fusion)



Extractive

It simply **takes out the important sentences** or phrases from the **original text** and joins them to form a summary.

How to choose the important sentences?

Ranking algorithm!

Assigns scores to each sentence in the text based on their relevance to the overall meaning of the document



TF - IDF Term Frequency-Inverse Document Frequency

Term Frequency

It is a measure of how frequently a term, t , appears in a document, d :

$$tf_{t,d} = \frac{n_{t,d}}{\text{Number of terms in the document}}$$

Example: "This movie is not scary and is slow"

Vocabulary: 'This', 'movie', 'is', 'very', 'scary', 'and', 'long', 'not', 'slow', 'spooky', 'good'

1. 1. 2. 0. 1. 1. 0. 1. 1. 0. 0

TF:	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{2}{8}$	0	$\frac{1}{8}$	$\frac{1}{8}$	0	$\frac{1}{8}$	$\frac{1}{8}$	0	0
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TF - IDF Term Frequency-Inverse Document Frequency

Inverse Document Frequency (IDF)

IDF is a measure of how important a term is.

$$idf_t = \log \frac{\text{number of documents}}{\text{number of documents with term 't'}}$$

Example:

Review 1: This movie is very scary and long

Review 2: This movie is not scary and is slow

Review 3: This movie is spooky and good

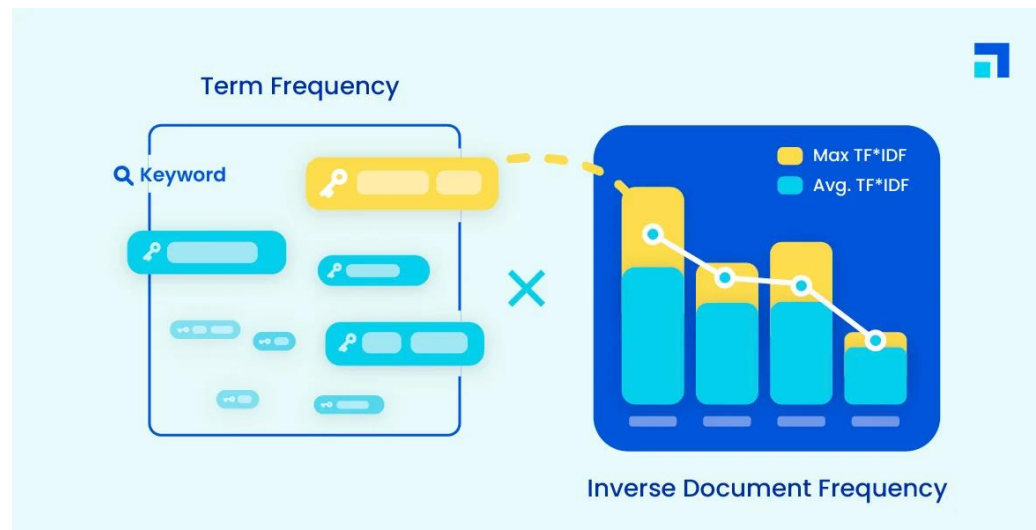
For Review 2: 'This', 'movie', 'is', 'not', 'scary', 'and', 'is', 'slow',
3. 3. 3. 1. 2. 3. 1

IDF: $\log \frac{3}{3}$ $\log \frac{3}{3}$ $\log \frac{3}{3}$ $\log \frac{3}{1}$ $\log \frac{3}{2}$ $\log \frac{3}{3}$ $\log \frac{3}{1}$

IDF ↑

Term	Review 1	Review 2	Review 3	IDF
This	1	1	1	0.00
movie	1	1	1	0.00
is	1	2	1	0.00
very	1	0	0	0.48
scary	1	1	0	0.18
and	1	1	1	0.00
long	1	0	0	0.48
not	0	1	0	0.48
slow	0	1	0	0.48
spooky	0	0	1	0.48
good	0	0	1	0.48

TF - IDF Term Frequency-Inverse Document Frequency



$$(tf_idf)_{t,d} = tf_{t,d} * idf_t$$

For Review 2: 'This', 'movie', 'is', 'not', 'scary', 'and', 'slow',

TF:	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{2}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$
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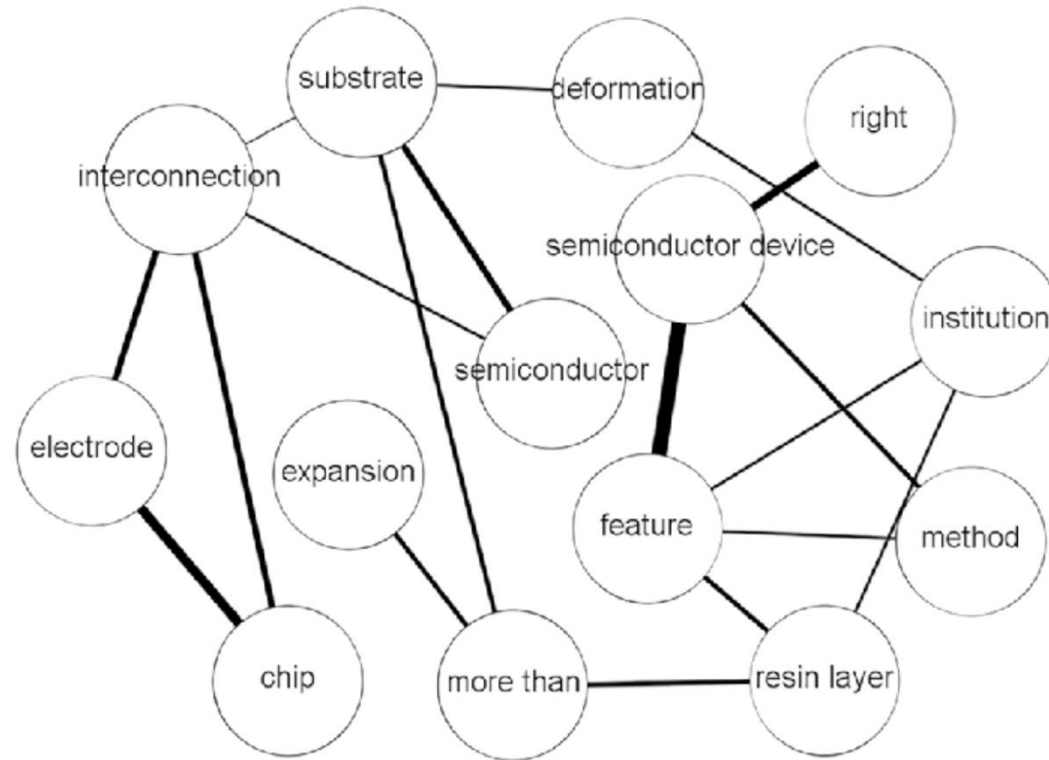
*

IDF:	$\log \frac{3}{3}$	$\log \frac{3}{3}$	$\log \frac{3}{3}$	$\log \frac{3}{1}$	$\log \frac{3}{2}$	$\log \frac{3}{3}$	$\log \frac{3}{1}$
------	--------------------	--------------------	--------------------	--------------------	--------------------	--------------------	--------------------

	0	0	0	0.06	0.023	0	0
---	---	---	---	------	-------	---	---

TextRank (graph-based)

- Sentences as nodes
- Edge weight = similarity
- **PageRank** selects central sentences
- Often reduces redundancy vs pure TF-IDF



TextRank (graph-based)

PageRank

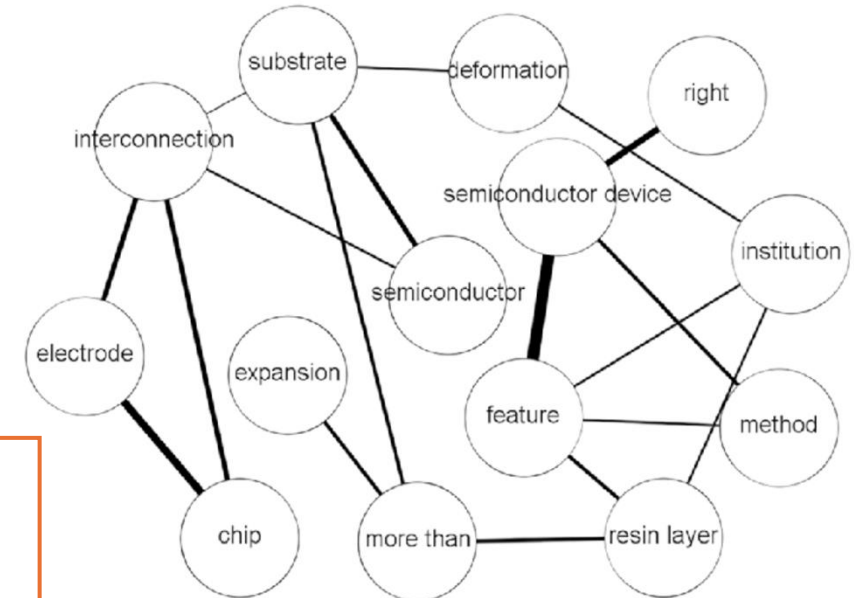
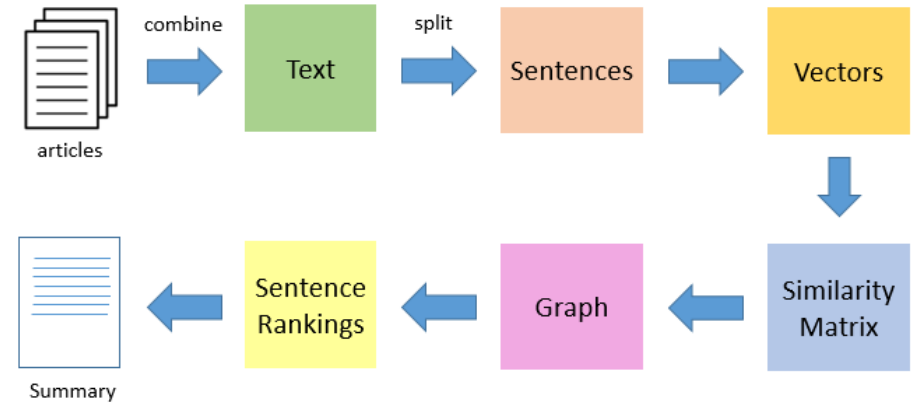
- Suppose we have 4 web pages.
- To capture the probabilities of users navigating from one page to another, create the matrix M
- Each element of this matrix denotes the probability of a user transitioning from one web page to another.



TextRank

- In place of web pages, we use sentences
- **Similarity between any two sentences** is used as an equivalent to the web page transition probability
- The **similarity scores** are stored in a square matrix, similar to the matrix M used for PageRank

If the sentence is similar to many key sentences, it receives many weighted votes and gets high rank → likely selected into summary.



Abstractive summarisation

- Model generates a **new summary** from the source
- Can **paraphrase** and **merge** information

Better readability, but riskier

ChatGPT primarily does *abstractive* summarisation ?

- **Prompt for Extractive:**

Highlight the most important sentences from the text without rewriting.

- **Prompt for Abstractive:**

Summarize the text in 3 sentences in your own words.

Evaluation summarisation

Why evaluation is hard for summarisation

- Many valid summaries
- Lexical overlap $\neq$ meaning
- **Faithfulness matters more than style**

A summary can be: very fluent, nicely structured, concise, and still be **factually wrong**. → **faithfulness** (is it supported by source?) is often **more important than style**.

- Source :
“The method improved **throughput** by 2–4× on tested **workloads**.”
- Bad summary: 
“The method consistently improves **accuracy** by 2–4× across **tasks**.”

Human / verifier-based methods matter

That's why summarisation evaluation often combines:

- ✓ **ROUGE** (content overlap)
- ✓ **semantic similarity**
- ✓ **faithfulness checks** (QA/NLI/LLM-judge)

QA based check

“does summary answer same questions as source?”

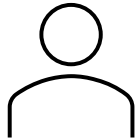
- ✓ **human rubric** (coverage, faithfulness, clarity)

Introduction Today

1. What “translation” means for LLMs
2. Prompting patterns that work
3. When fine-tuning wins (and why)

What do we mean by “translation” here?

- **Language** ↔ language
- **Style/reading** level/domain adaptation
- **Structured** ↔ structured (JSON, schema, code)



“Please submit the reimbursement form by Friday. Keep all receipts.”

formal

“Patients presenting with hypertension should adhere to a sodium-restricted diet to mitigate cardiovascular risk.”

Compute moving average over the last N points
$$\text{avg} = \text{sum}(x[-N:]) / N$$



French

Veuillez soumettre le formulaire de remboursement d'ici vendredi. Conservez tous les reçus.

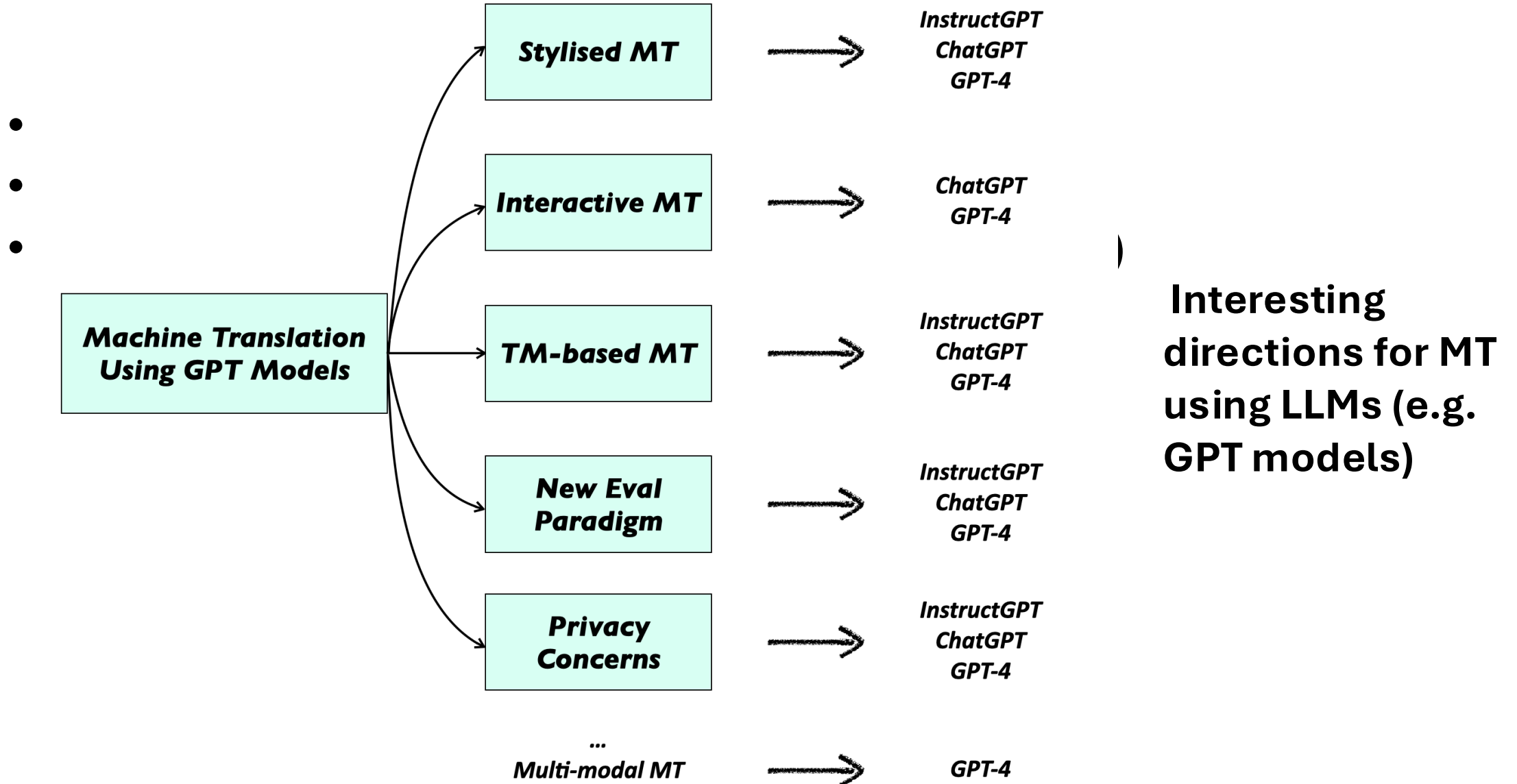
Plain

If you have high blood pressure, eat less salt. This can help protect your heart.

Chinese

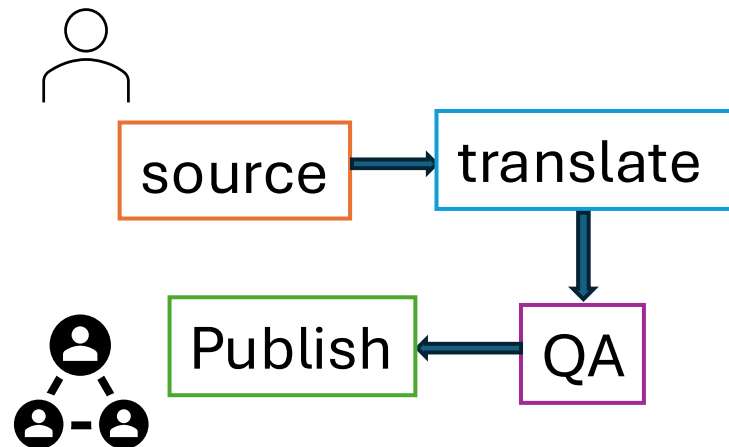
计算最近 N 个点的移动平均值
$$\text{avg} = \text{sum}(x[-N:]) / N$$

What do we mean by “translation” here?



Where LLM translation is used (real workflows)

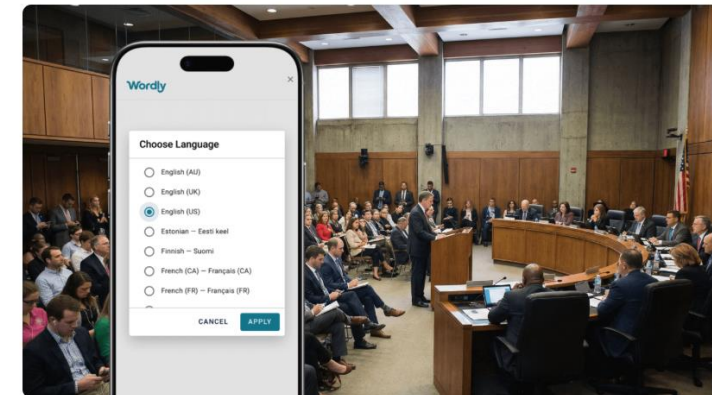
- Customer support,
- Legal/compliance,
- Product localization,
- Medical notes



Live AI Translation & Captions

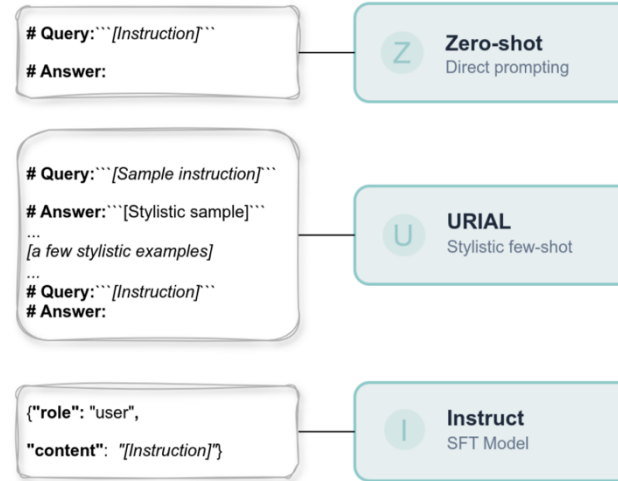
Government Translation Services

Make government meetings more inclusive, accessible, and engaging with live translation and captions powered by AI. Wordly makes it easy and affordable to offer audio translation, captions & subtitles, transcripts, and summaries for all of your government meetings and events.

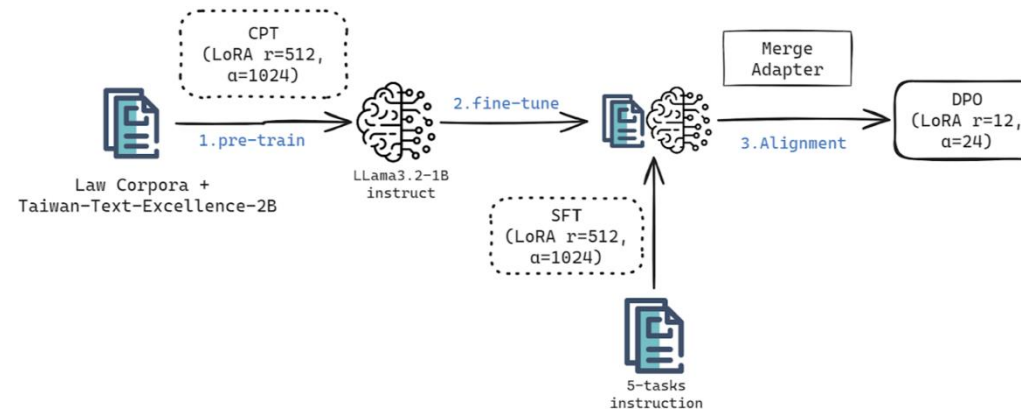


Two families of approaches

Prompting / In-context learning

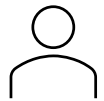


Finetuning (SFT)



Baseline: Zero-shot prompt

- Zero-shot translation is when you ask an LLM to translate **without giving any in-prompt examples** and without any fine-tuning for that specific translation task.
- You just give an instruction like “Translate X to Y” and the model relies on its general pretrained multilingual ability.



Translate the following English text into French:
“The system will be unavailable from 2:00 to 3:30 PM for scheduled maintenance.”

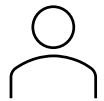


« Le système sera indisponible de 14 h 00 à 15 h 30 en raison d’une maintenance planifiée. »

“zero-shot with constraints” (still zero-shot)

Baseline: Zero-shot prompt

- Zero-shot translation = “Translate” instruction + input text, with **no in-prompt examples (and no fine-tuning)**.
- You just give an instruction like “Translate X to Y” and the model relies on its general pretrained multilingual ability.

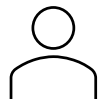


Translate the following English text into French:
“The system will be unavailable from 2:00 to 3:30 PM for scheduled maintenance.”



« Le système sera indisponible de 14 h 00 à 15 h 30 en raison d’une maintenance planifiée. »

“zero-shot with constraints” (still zero-shot)



Translate into Spanish.

Constraints: keep numbers and times exactly, keep product name “DataHub” unchanged.

Text: “DataHub will be unavailable from 2:00 to 3:30 PM.”

Baseline: Zero-shot prompt

- **Simplest prompt**

You give the most basic instruction, e.g. *“Translate this text from English to French.”* + the text. No examples, no glossary, no extra rules.

- **Surprisingly strong**

Modern LLMs often produce **fluent, generally correct** translations even with that minimal instruction—especially for common language pairs and everyday text.

- **Fragile on constraints**

The same minimal setup often **fails when you require strict rules**, like:

- keep **numbers/dates** exactly unchanged
- preserve **HTML/XML tags** exactly 

Baseline: Zero-shot prompt

- **Simplest prompt**

You give the most basic instruction, e.g. no examples, no glossary, no extra rules.

- **Surprisingly strong**

Modern LLMs often produce **fluent, good** instruction—especially for common language tasks.

- **Fragile on constraints**

The same minimal setup often **fails with**

- keep **numbers/dates** exactly unchanged
- preserve **HTML/XML tags** exactly

HTML tag break

extra/missing closing tag,
tag moved,
or punctuation inserted inside tags.

ChatGPT 5.2 ▾

Translate into chinese. Preserve the HTML exactly (all tags must remain exactly the same and correctly closed). Text: Click **Save** to continue. If you see **<i>Error 17</i>**, contact support.

点击 **保存** 以继续。如果您看到 **<i>Error 17</i>**，请联系支持。



Translate into Spanish. Preserve the HTML exactly (all tags must remain exactly the same and correctly closed).

Text:
Click **Save** to continue. If you see **<i>Error 17</i>**, contact support.

Haga clic en **Guardar** para continuar. Si ve **<i>Error 17</i>**, contacte con el soporte técnico.

Baseline: Zero-shot prompt

- **Simplest prompt**

You give the most basic instruction, e.g. *“Translate this text from English to French:”* + the text. No examples, no glossary, no extra rules.

- **Surprisingly strong**

Modern LLMs often produce **fluent, generally correct** translations even with that minimal instruction—especially for common language pairs and everyday text.

- **Fragile on constraints**

The same minimal setup often **fails when you require strict rules**, like:

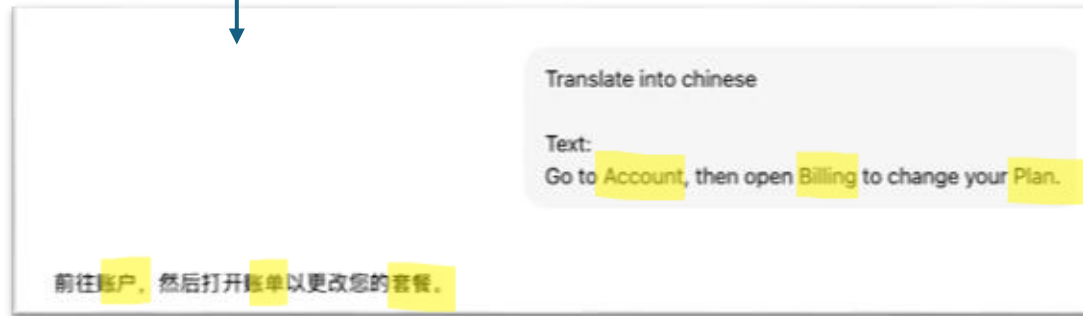
- keep **numbers/dates** exactly unchanged
- preserve **HTML/XML tags** exactly
- enforce a **glossary** (“‘bill’ must always be translated as ‘facture’ not ‘addition’”)
- produce **valid JSON** with unchanged keys
- keep **named entities** unchanged



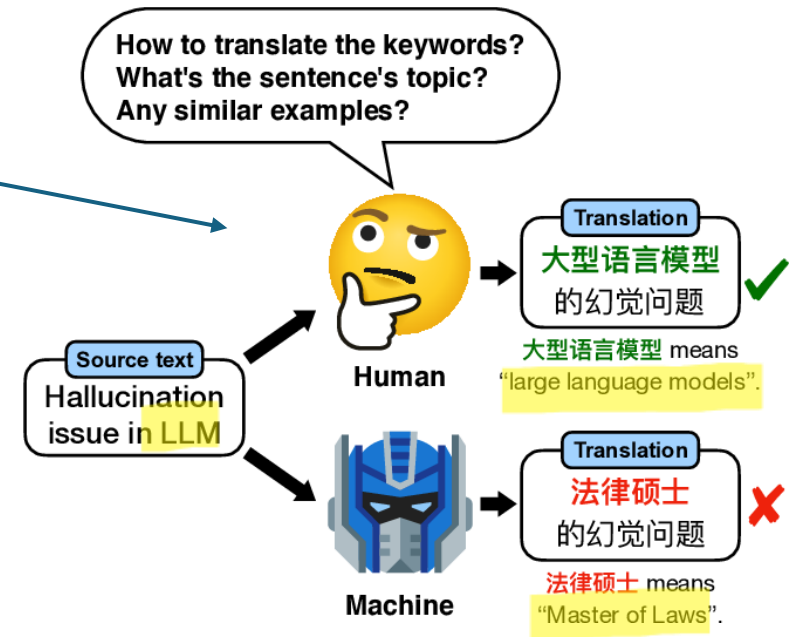
```
{"title":"Account settings","cta":"Save changes","error_invalid_email":"Please enter a valid email address."}
```

Why translation is hard for LLMs

- Ambiguity & polysemy: same word, different sense
- Terminology consistency: glossary drift across a document
- Formatting constraints (tags, numbers, entities)

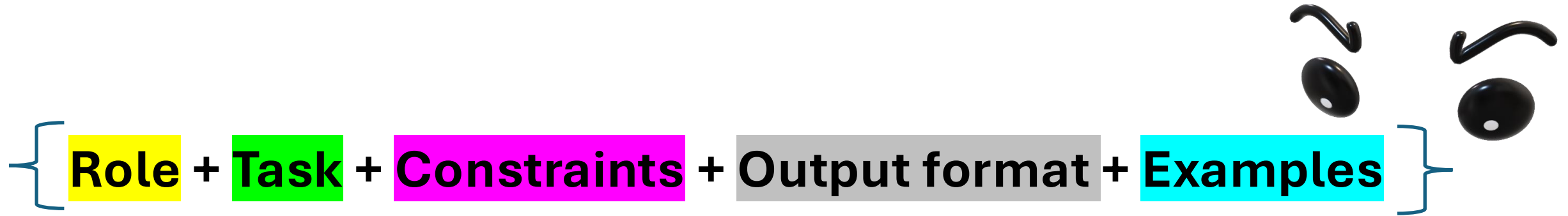


Glossary: “token” must stay **token** (not “jeton”) in this course context
Source: “Token limits vary by model.”
Wrong (FR): “Les **jetons** ...” (may be unacceptable in this domain)
Correct (FR): “Les **tokens** ...”



[Exploring Human-Like Translation Strategy with Large Language Models](#)

A minimal “production” prompt template



Prompting pattern 1: “constraints-first” translation

- Do not change numbers/names
- Preserve markup
- Keep glossary terms fixed

Prompting pattern

Prompting pattern 1: “constraints-first” translation

Prompting pattern 2: Few-shot with style + glossary

- include 1–2 examples;
- include glossary table;
- enforce tone

Source text: 链博会是“链”通世界的盛会，也是“链”接未来的盛举，展现了各方构建稳定和富有韧性的产业链供应链的共同心声。

You are a professional translator. ... These sentences will be displayed below.

Chinese text: 李强总理就深化产业链供应链国际合作提出四点倡议，倡导共同构筑安全稳定、畅通高效、开放包容、互利共赢的产业链供应链，希望各国企业家为推动全球产业链供应链稳健运行发挥积极作用。

Translation: Premier Li put forward a four-point proposal on deepening international cooperation on industrial and supply chains. He called for joint efforts to build a global industrial and supply chain system that is secure, stable, smooth, efficient, open, inclusive and mutually beneficial and expressed the hope that entrepreneurs around the world will play a positive role in safeguarding the sound operation of global industrial and supply chains. China will offer more facilitation and better services for foreign businesses investing or operating in China.

After the example pairs, ... Translate the text from Chinese to English.

You are a professional translator. ... These sentences will be displayed below.

Chinese text: 王毅外长在会上表示，自第七次外长会以来，各方秉持“发展为先、平等协商、务实高效、开放包容”的澜湄精神，扎实推进各领域合作，取得系列新成果，打造了多个新亮点。

Translation: Foreign Minister Wang Yi noted, since the seventh LMC Foreign Ministers' Meeting, relevant parties have acted in the Lancang-Mekong spirit of development first, equal consultation, practicality and high efficiency, and openness and inclusiveness, taken steady steps to advance cooperation in various areas, achieved a number of new outcomes and created multiple new highlights.

After the example pairs, ... Translate the text from Chinese to English.

Prompting pattern

Prompting pattern 1: “constraints-first” translation

Prompting pattern 2: Few-shot with style + glossary

Prompting pattern 3: Structured output (JSON / XML)

- Output schema
- Validation mindset
- Retries on schema violation

Translate ONLY the JSON string values into German.

Rules:

- 1) DO NOT change any JSON keys.
- 2) Keep {placeholders} exactly unchanged.
- 3) Output must be valid JSON (no extra text).

Input:

```
{"title": "Account settings", "cta": "Save changes", "msg": "Hi {first_name}, update your email.", "error_invalid_email": "Please enter a valid email address."}
```

```
{"title": "Kontoeinstellungen", "cta": "Änderungen speichern", "msg": "Hallo {first_name}, aktualisieren Sie Ihre E-Mail-Adresse.", "error_invalid_email": "Bitte geben Sie eine gültige E-Mail-Adresse ein."}
```

Prompting pattern

Prompting pattern 1: “constraints-first” translation

Prompting pattern 2: Few-shot with style + glossary

Prompting pattern 3: Structured output (JSON / XML)

Prompting pattern 4: Two-pass translate → revise

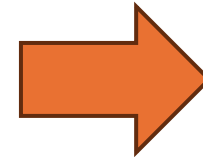
- Draft translation
- Revision pass: terminology + formatting + tone

Step 1

Translate English to German:

"Hi {first_name}, your order #{order_id} is ready. Click **Track** to see delivery updates."

„Hallo {first_name}, Ihre Bestellung #{order_id} ist bereit. Klicken Sie auf **Verfolgen**, um Lieferupdates zu sehen.“



Step 2

Revise the German translation below.

Rules (must follow):

- 1) Keep {first_name} and {order_id} EXACTLY unchanged.
- 2) Keep the HTML tags exactly: **Track** must remain **Track** (do not translate Track, do not change tags).
- 3) Do not add or remove information.
- 4) Keep a friendly customer-support tone.

German draft:

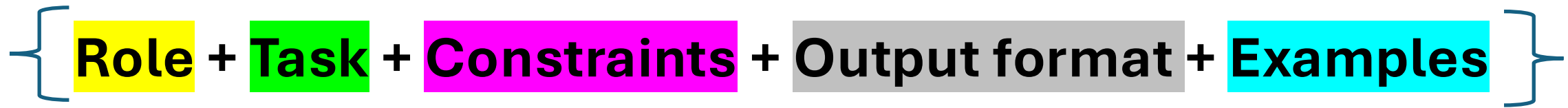
Hi {first_name}, your order #{order_id} is ready. Click **Track** to see delivery updates.

Return only the revised German sentence.



Hallo {first_name}, Ihre Bestellung #{order_id} ist bereit. Klicken Sie auf **Track**, um Lieferupdates zu sehen.

Prompting pattern



Prompting pattern 1: “constraints-first” translation

Prompting pattern 2: Few-shot with style + glossary

Prompting pattern 3: Structured output (JSON / XML)

Prompting pattern 4: Two-pass translate → revise

Prompting pattern 5: Self-check (targeted)

- List changed entities
- List changed numbers
- Glossary compliance: yes/no



Common prompting failure modes

- Over-literal → add style instruction + example
- Hallucinated additions → “no new info” + self-check
- Glossary drift → glossary + forced mapping

When prompting is enough?

Common prompting failure modes

- Over-literal → add style instruction + example
- Hallucinated additions → “no new info” + self-check
- Glossary drift → glossary + forced mapping

When prompting is enough?

low volume, flexible style, no strict terminology, rapid iteration

Prompting Cost

Total cost per doc = **Model cost** + **Engineering overhead** + **Human QA cost**

Model cost

= (input_tokens + output_tokens) ÷ 1,000 × price_per_1k_tokens × #calls

Latency cost (time) *end-to-end time until the final token*

= #calls × avg_latency_per_call (seconds) (+ retries if you use validation)

Human QA cost

= QA_minutes_per_doc × QA_rate_per_minute

Why fine-tuning can beat prompts (conceptual)

- Internalizes **consistent style**/terminology
 - ✓ Fine-tuning bakes the mapping into the model so you don't need to restate glossary rules every time—and consistency improves across many short strings.
- Better adherence under **long context**
 - ✓ Fine-tuning doesn't “guarantee” perfection, but it reduces *late-document drift* because consistency is learned rather than carried entirely in prompt memory.
- **Lower prompt** overhead
 - ✓ With fine-tuning, you “pay once” in training effort and then save repeatedly in inference prompt length.

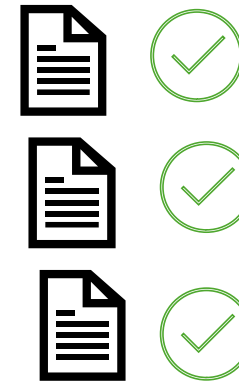
Input: 6 page document contains 5 sections



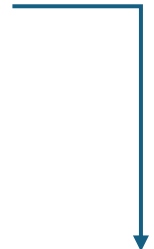
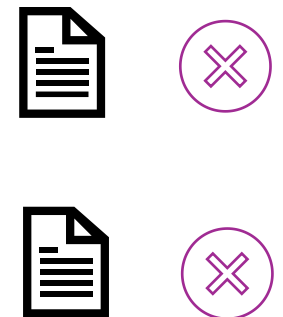
Translate

1. Keep defined term consist
2. Preserve numbering
3.

Section 1- 3

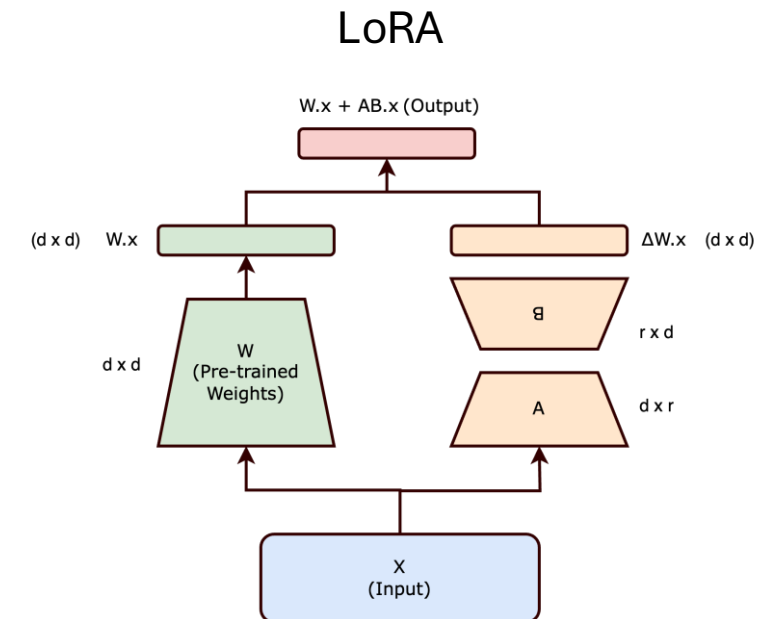
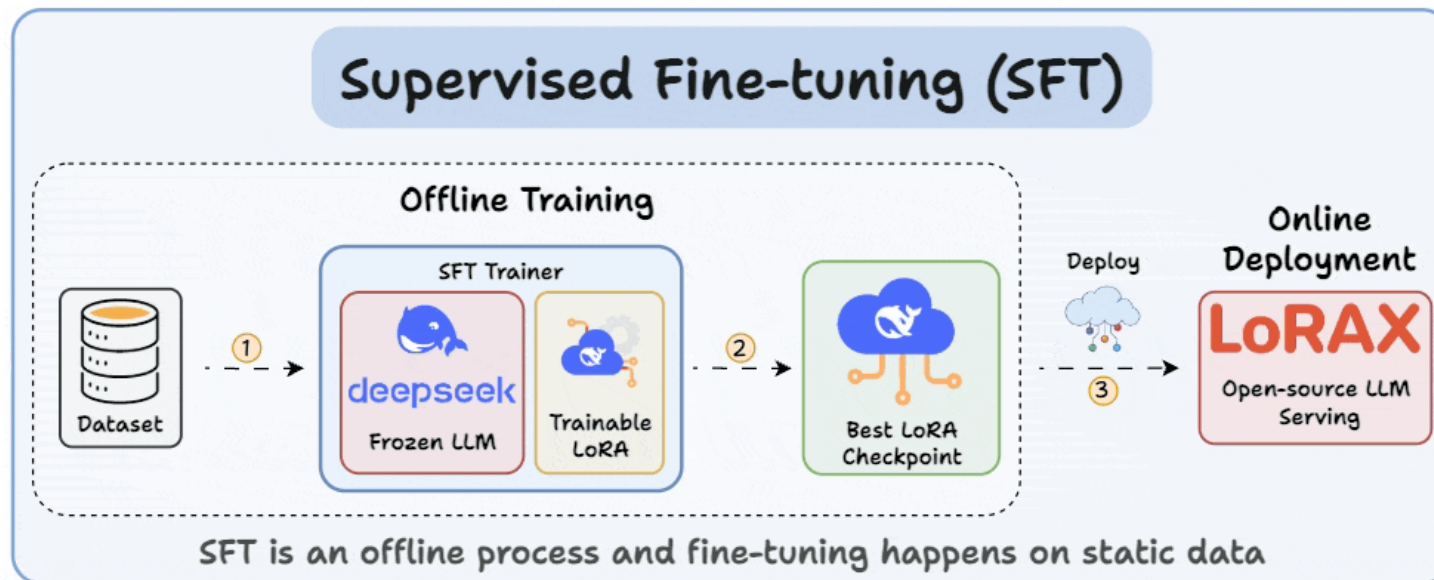


Section 4 & 5



Finetuning - Recall

- SFT (supervised finetuning),
- Instruction tuning,
- Adapters/LoRA



What data you need

- Same intent, no additions/omissions
- Placeholders/tags preserved exactly
- Glossary terms consistently enforced
- Metadata (tone, locale): tone consistent across examples

```
{
  "id": "ui_001",
  "source_lang": "en",
  "target_lang": "de",
  "domain": "ui",
  "tone": "neutral",
  "constraints": ["placeholders"],
  "source_text": "Hi {first_name}, your order #{order_id} is ready.",
  "target_text": "Hallo {first_name}, Ihre Bestellung Nr. {order_id} ist bereit."
}
```

Bad Data

- **Noisy alignments (wrong target paired to source)**

Source: "Reset your password."

Target: "Votre compte est verrouillé." (different meaning)

- **Mixed style labels / inconsistent tone**

"Your payment failed." → "Le paiement a échoué." (neutral)

"Your payment failed." → "Oups ! Votre paiement n'est pas passé 😊" (casual emoji)

- **Inconsistent glossary in the training targets**

- **Format violations inside gold data**

Source: Hi {first_name}

Target: Bonjour {prénom} ❌ (placeholder changed)

Source HTML: Save

Target: Enregistrer ❌ (invalid tags)

- **Privacy/IP leakage**

includes real emails, phone numbers, customer IDs, internal URLs

Outputs unpredictable!

Demo: Fine-tuning an MT model via SFT

Setup

- ✓ Model: Helsinki-NLP/opus-mt-en-de (small, fast)
- ✓ Method: LoRA (parameter-efficient)
- ✓ Packages: transformers, datasets, evaluate, peft, accelerate

```
from datasets import load_dataset
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM

model_name = "Helsinki-NLP/opus-mt-en-de"
tokenizer = AutoTokenizer.from_pretrained(model_name)
base_model = AutoModelForSeq2SeqLM.from_pretrained(model_name)
```

Dataset

- 60–200 examples is enough to demo behavior change
- Include “hard cases”:
 - {placeholders} unchanged
 - UI terms fixed (“Account”→“Konto”)
 - HTML tags preserved
- Split: train / validation / test (e.g., 80/10/10)

- {"src":"Hi {first_name} ...","tgt":"Hallo {first_name} ...","type":"placeholder"}
- {"src":"Click Save","tgt":"Klicken Sie auf Save","type":"html"}

Preprocessing (tokenization + labels)

- Tokenize **src** as encoder **input**
- Tokenize **tgt** as decoder **labels**

```
model_inputs = tokenizer(
    batch["src"], max_length=max_source_len, truncation=True
)
with tokenizer.as_target_tokenizer():
    labels = tokenizer(
        batch["tgt"], max_length=max_target_len, truncation=True
    )["input_ids"]
```

Demo: Fine-tuning an MT model via SFT

Training

Trainer computes loss automatically for seq2seq:
teacher forcing: model predicts next target token
cross-entropy loss sums over target tokens

Objective: Maximize{ $P(\text{target_text} \mid \text{instruction, source_text})$ }

LoRA

```
from peft import LoraConfig, get_peft_model, TaskType
from transformers import DataCollatorForSeq2Seq,
                        Seq2SeqTrainingArguments, Seq2SeqTrainer

lora_config = LoraConfig(
    task_type=TaskType.SEQ_2_SEQ_LM,
    r=8,
    lora_alpha=16,
    lora_dropout=0.05,
    target_modules=["q", "v"]
)

model = get_peft_model(base_model, lora_config)
model.print_trainable_parameters()
```

```
pred_text = tokenizer.batch_decode(preds, skip_special_tokens=True)
label_text = tokenizer.batch_decode(labels, skip_special_tokens=True)

args = Seq2SeqTrainingArguments(
    output_dir="mt_lora_demo",
    learning_rate=2e-4,
    per_device_train_batch_size=8,
    per_device_eval_batch_size=8,
    num_train_epochs=8,
    evaluation_strategy="epoch",
    save_strategy="no",
    predict_with_generate=True,
    logging_steps=5,
    report_to=[])
```

```
collator = DataCollatorForSeq2Seq(tokenizer, model=model)
trainer = Seq2SeqTrainer(
    model=model,
    args=args,
    train_dataset=tokenized["train"],
    eval_dataset=tokenized["validation"],
    tokenizer=tokenizer,
    data_collator=collator, (padding)
    compute_metrics=compute_metrics (Evaluate)
)
trainer.train()
```

Evaluation

- **Automatic**

BLEU score(word n-gram overlap); *chrf* (character n-grams)

- **Human rubric**

Adequacy (meaning preserved)

Terminology/Consistency

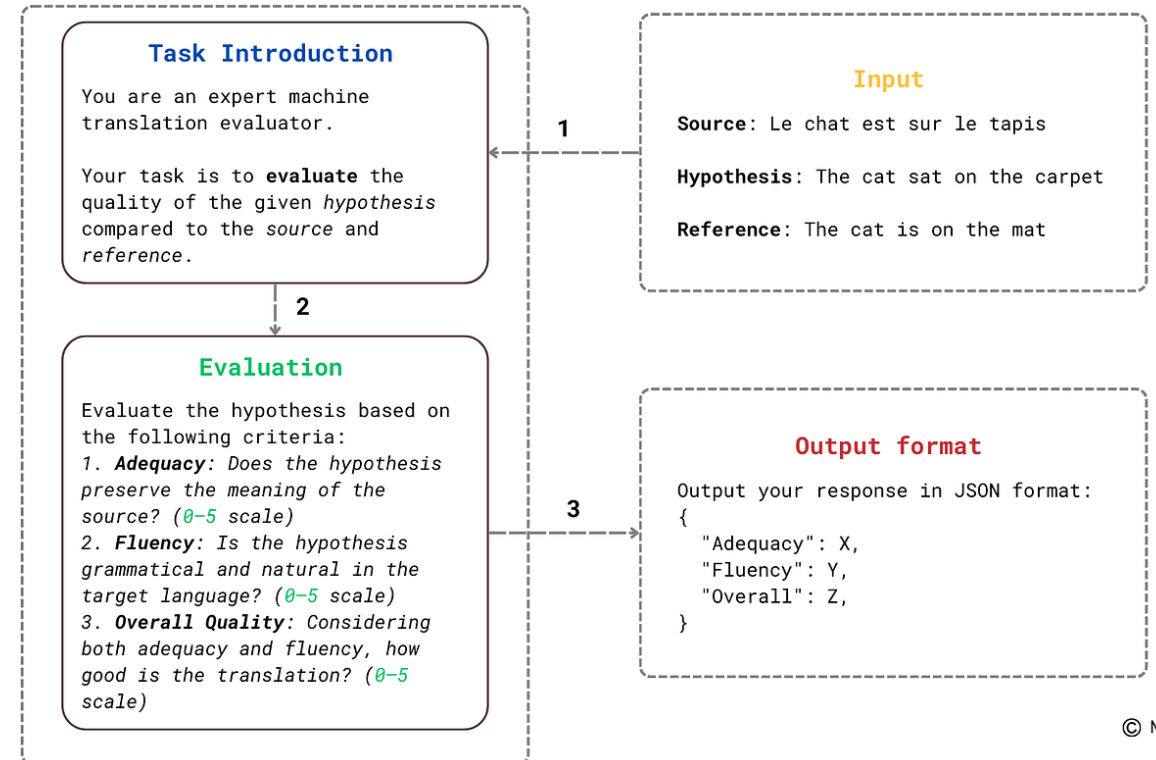
Format/Constraints:

- **Constraint checks**

Exact –match check (place holder/tags...)

Glossary check

Length / structure check



Evaluation- chrF (optional)

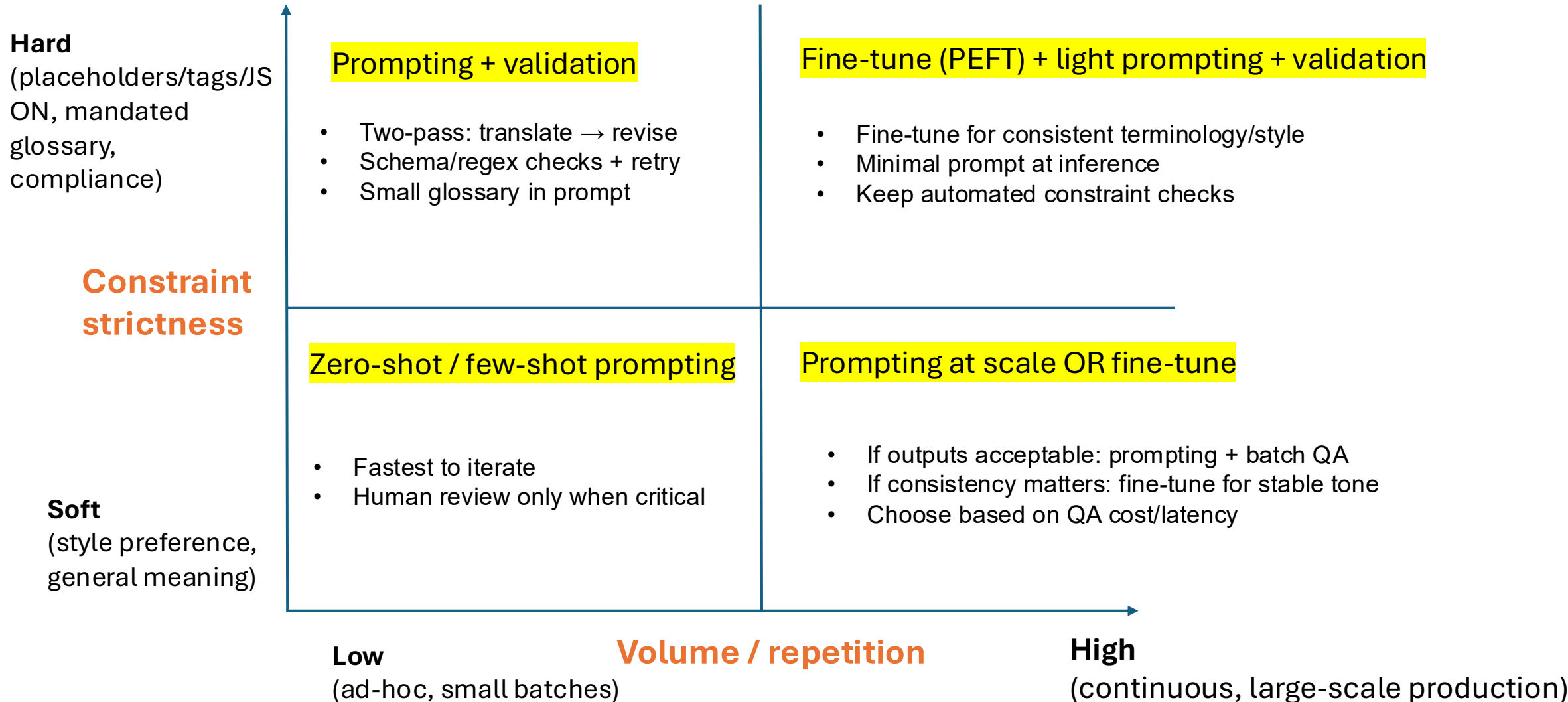
- Break both **hypothesis** and **reference** into **character n-grams** (e.g., for $n=1\dots6$).
- Compute **precision** (how many character n-grams in the hypothesis also appear in the reference) and **recall** (how many reference n-grams are covered).
- Combine them with an **F-score** (usually with recall weighted a bit more than precision).

Metric	Reference	Hypothesis	Match Type
BLEU	The cat is on the mat	The cat sat on the carpet	Exact word sequence matches
METEOR		The cat sat on the carpet	Flexible word and synonym matches
ChrF		The cats at on the carpet	Partial matches within words
TER		The cat is on the mat	Number of edits (insertions, deletions, substitutions, shifts) needed

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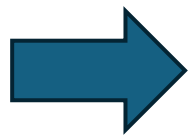
[chrF: character n-gram F-score for automatic MT evaluation](#)

Prompting vs Fine-tuning



Take Away

- Use prompting for speed/experimentation
- Fine-tune for consistency at scale
- Always evaluate constraints explicitly



Translation often needs **external knowledge**: terminology
DBs, product names, policies → retrieval

RAG

Next steps

- RAG: Retrieval-Augmented Generation

Additional Material:

- [In-context Learning vs. Instruction Tuning: The Case of Small and Multilingual Language Models](#)
- [Prompting Large Language Model for Machine Translation: A Case Study](#)
- [Finetuned Language Models Are Zero-Shot Learners](#)
- [chrF: character n-gram F-score for automatic MT evaluation](#)

Thank you!

If you want to ask us any questions about the module, the university, or anything else, drop us an email.

Pro tip: write **COMM117** on the email subject line, to make sure we give it priority.



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Questions?